Full paper outline

Practical Incentive-Compatible Drafts: Carry-Over Lottery Allocation

MIT Sloan Sports Analytics Conference 2027 (SSAC27), Basketball track, Paper ID 100

Draft for discussion, 3 November 2026

This outline works backwards from the abstract SSAC accepted, which presents Carry-Over Lottery Allocation (COLA), a family of draft rules for the National Basketball Association (NBA). The paper is due at 11:59 PM EST on Tuesday, 1 December 2026, in SSAC's Word template. The template sets no page limit and notes that papers typically run under 8 pages, so the outline budgets about 4,000 words for the main text and moves the proofs and full tables to an appendix. Each section ends with what our current manuscript, `main.tex`, already has for it. All of it is open to comments and edits.

§	Section	Words	Abstract promise it keeps
1	Introduction	500	The dilemma, the reframe, the contributions
2	Background	350	Munro and Banchio's impossibility result; 3-2-1
3	The COLA framework	600	Three dials; 3-2-1 as an edge case
4	Incentive compatibility	550	The proof and the one assumption that needs justification
5	When the assumptions fail	450	"The impact of violating model assumptions … and how to mitigate issues"
6	Simulation design	450	Nine mechanisms, 30 leagues, 25 seasons, paired intervals
7	Results	700	Table 1; other starting positions; six more mechanisms
8	Discussion and adoption	300	"One example of many"; auction credit
9	Conclusion	100	The dilemma resolved
	Main text	4,000	

Sections

1. Introduction (500 words)

- The tradeoff between helping weak teams and the tanking it invites, and the NBA's lottery changes up to 3-2-1.
- The reframe, weakness measured by multi-season playoff results, under which losing more games buys nothing.
- The contributions: the COLA framework with 3-2-1 inside it, an incentive-compatibility proof under stated assumptions, nine mechanisms simulated on the same leagues, and Simple COLA as a one-line rule with no lottery.

In `main.tex`. §1 has the lottery's history through 3-2-1, the evidence of tanking, Munro and Banchio's result and the reframe. Its contributions list centers on Waitlist COLA, so it needs rewriting around Simple COLA.

2. Background (350 words)

- Evidence that teams tank.
- Munro and Banchio's result that any mechanism based on end-of-season records can reward losing.
- 3-2-1, adopted on 28 May 2026, and the other proposals (the uniform lottery, the wheel, the Gold plan and auction drafts).

In `main.tex`. §2 covers all three in 646 words, except the auction proposals, which are in §6.2.

3. The COLA framework (600 words)

- The waiting number and the three dials, which set how priority increases, how it decreases and how it maps to picks. The seven-dial parameterization moves to Appendix B.
- A one-line definition of each of the nine rules.
- 3-2-1 as an edge case, where record sets priority and the only carry-over is the bar on repeat top picks. It still pays for losing from the play-in positions, by 0.37 places from eleventh-worst and 2.02 from fourteenth-worst.

In `main.tex`. §3.1–3.3 define the index, the seven dials and six of the nine rules, §5.2 describes the other three, and §3.6 places 3-2-1 in the space. Prof. Highley's note in §1 already groups the seven dials into four questions, and the abstract asks three of them. Its definition of Simple COLA resets only on a series win, so the top-3 pick needs adding, and the play-in figures are new.

4. Incentive compatibility (550 words)

- The theorem for Simple COLA and its assumptions, with a proof sketch in the text and the full proof in Appendix A.
- The one assumption that needs justification, that teams prefer winning games to manipulating the pool of lottery teams.

- On our runs, losing never gained a team a pick under Simple COLA across 158,325 counterfactuals.

In `main.tex`. §3.5 lists the five assumptions under which the original COLA paper (arXiv 2602.02487) proves incentive compatibility for Classic COLA, and §4.6 gives three ways to close the pool assumption. The theorem and proof for Simple COLA are new, and the counterfactual follows the method of Table 5.

5. When the assumptions fail (450 words)

- The incentive to stay eligible, priced as the draft places the longest-waiting team gains on the lottery side of the boundary: 1.4 under the old lottery, 5.0 under 3-2-1, 9.6 under Full-lottery COLA and 22 under Simple COLA, whose boundary is a first-round loss.
- Longer rebuilds after a reset, measured as the next season's pick for a team that took a top-3 pick and missed the playoffs again: 6.6 under the old lottery, 8.6 under 3-2-1, 9.4 under Full-lottery COLA and 21.0 under Simple COLA.
- Draft-class strength, which none of the rules condition on.
- Mitigations and what each adds in complexity: a cap on the waiting number, a reset on playoff appearance, and a lottery over the top picks.

In `main.tex`. §4.2 bounds the gain from pool manipulation (Theorem 2), §4.5 bounds the gain from losing a playoff series under Capped COLA (Lemma 4), which is the starting point for pricing a cap, and §4.3 raises draft-class strength as the pick-value gap. The eligibility and rebuild measurements are new.

6. Simulation design (450 words)

- ZenGM Basketball-GM, 30 leagues of 25 seasons on shared seeds, and paired differences with 95% intervals.
- Four measures across the three criteria: help for a worse record, help for a longer wait, the longest wait for a series win or top-3 pick, and the average wait for the next series win.
- Validation against the NBA over 2001-02 to 2025-26. The simulation matches the average playoff drought (2.89 seasons against 2.99), the wait for the next series win (4.65 against 4.55) and the year-to-year correlation of win percentage (0.62 against 0.60).

In `main.tex`. §5.1 describes the engine and why it suits the question. Its runs use 48 leagues of 15 seasons, scored differently, so the run design, three of the four measures and the comparison with the NBA are new.

7. Results (700 words)

- A table of all nine rules on the four measures, with paired intervals against 3-2-1.
- A table of the reward for losing by starting position. From fifth-worst it is $+1.32$ places under the old lottery, -0.63 under 3-2-1 and -0.94 under Simple COLA; from fourteenth-worst it is $+9.96$, $+2.02$ and -3.50.
- Help, where Simple COLA cuts the longest wait for a series win or top-3 pick from 16.3 seasons under 3-2-1 to 5.5.
- Parity, where the rules are level on the wait for the next series win. Droughts end 80 to 87 times per league across the nine rules and 88 times in the NBA over the same 25 seasons, so the draft changes which teams end their droughts and barely changes how often droughts end.
- Simple COLA's one parity cost, a wait to return to the playoffs 0.18 seasons longer than 3-2-1's (95% interval 0.08 to 0.28), because 39% of its top-3 picks go to teams that lost in the first round.

In `main.tex`. Table 5 already prices the reward by starting position for the old lottery, and §5.3 already finds parity level across rules, since a first-overall pick is worth about two to three wins. The nine-rule table and every figure for 3-2-1 and Simple COLA are new.

8. Discussion and adoption (300 words)

- Choosing a rule from the family, with Simple COLA for the most help and a one-line rule, and Full-lottery COLA where parity matters more or a lottery night is wanted.
- 3-2-1 as a stopgap, with COLA as the next step.
- Auction drafts, where carry-over credits that grow with waiting and reset with success allocate the currency without rewarding a worse record.
- Limitations: one simulator, tanking measured by counterfactual, and draft-class strength.

In `main.tex`. §6.1 reads 3-2-1 as a stopgap, §6.2 makes the auction argument in 682 words, and §6.3 sorts the dials by audience. The limitations in §5.3 need updating to these runs.

9. Conclusion (100 words)

The abstract's Conclusion, extended to the whole family.

In `main.tex`. The closing paragraph of §6 makes the same core argument for Waitlist COLA.

Appendix

A. The incentive-compatibility proof for Simple COLA.
B. The seven dials and the definition of every rule.
C. The pool-manipulation bounds.
D. Full results tables with intervals.
E. The validation table against the NBA.
F. The repository and the script behind each number.

In `main.tex`. B is §3.2–3.3 with Table 1, C is §4.1–4.4, and the Reproducibility paragraph in §6 starts F.

Open questions

1. Should anything else from `main.tex` stay? The largest parts the outline leaves out are its lead result (the decomposition of the old lottery's reward for losing), the 48-league evaluation built around Waitlist COLA, the historical backtest and the reset-on-championship discussion.
2. How should §5 treat a team's incentive to stay eligible for the lottery? Under Simple COLA, the team with the longest wait gains about 22 draft places by losing a first-round series. The paper can fix this with a cap on the waiting number or a reset on playoff appearance, tested in the simulator, or it can disclose the incentive.
3. How should the paper word 3-2-1's remaining reward for losing? The abstract says 3-2-1 has no reward for losing, and §3 reports one from the play-in positions.